

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

1. Code Layout Checklist Table

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	The project follows consistent Python PEP 8 standards and clean HTML/CSS indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	Source code is organized into templates, static, model, dataset, and deployment directories.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive, self-documenting names are used for variables, datasets, model metrics, and input fields.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly describe their purpose, such as preprocessing, training, prediction, and validation.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections of the code include comments for better understanding and maintenance.
6	Modular Design	Code is split into reusable functions/modules	Yes	The project is divided into reusable modules for preprocessing, model loading, training, and prediction.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated logic is minimized by using reusable functions and utility modules.

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation, model loading checks, and exception handling are implemented to improve reliability.

2. Reusable Components / Modules Table

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Preprocessing functions	Python	Data cleaning, categorical encoding, and feature standardization	High
2	Model loading scripts	Python	Loading serialized model parameters (rdf.pkl) inside the Flask app	High
3	ML Training Pipelines	Python	Dataset balancing (SMOTE) and classifier model comparison	High
4	Frontend Base Templates	HTML, CSS, Jinja2	Shared layout elements for the main interface and analytics dashboard	High
5	Client-Side Validators	JavaScript	Real-time input boundary validation and dynamic Lakhs calculations	Medium

3. Overall Code Quality Assessment Table

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized folder structure with clear separation of frontend, backend, and machine learning modules.
Readability	5	Code follows consistent naming conventions, indentation, and formatting, making it easy to understand.
Reusability	5	Modular components and reusable helper functions minimize duplication and simplify maintenance.
Documentation / Comments	4	Important logic is documented with comments, and the project includes README and supporting documentation.

Aspect	Rating (1–5)	Comments
Overall Score	4.8 / 5	The project demonstrates good coding practices, modular architecture, and maintainable design.

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